

Assignment 2 – Data Exploration

1. Data Cleaning

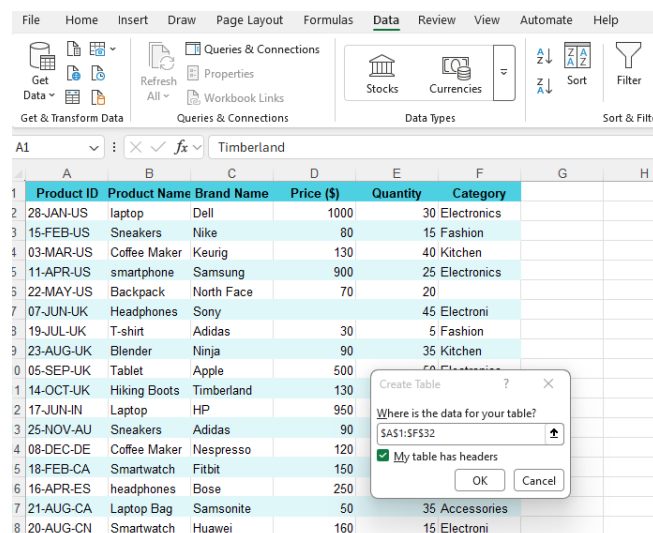
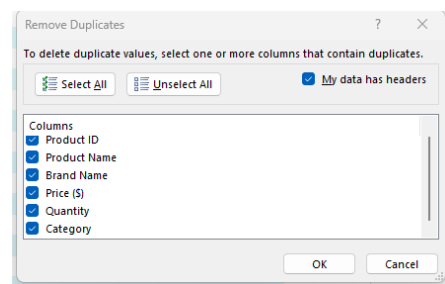
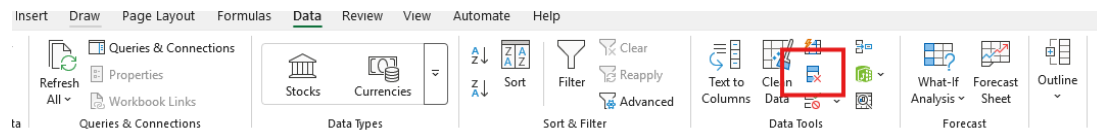
1.1 Removing Duplicates

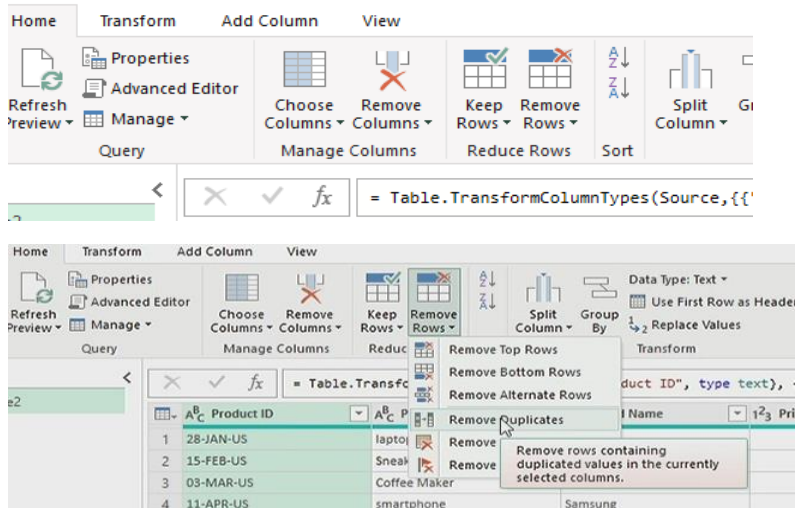
Datasets often contain duplicate records. To ensure accuracy, duplicates were identified and removed using the following fields:

- Product ID
- Product Name
- Brand Name
- Price
- Quantity
- Category

Excel's **Remove Duplicates** tool was used, and **3 duplicates were removed**. Duplicates were also removed using **Power Query** through:

- Remove Rows → Remove Duplicates





2. Correcting Inconsistent Data

Inconsistent values were found in the **Product Name** and **Category** columns.

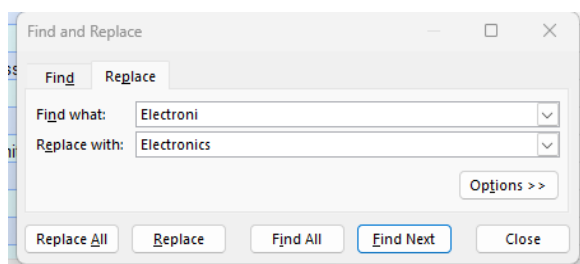
- identified that the Product Name column contains some unclean data

`=PROPER(TRIM(CLEAN([@[Product Name]])))`

`=PROPER(TRIM(CLEAN([@[Product Name]])))`

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Column1
28-JAN-US	laptop	Dell	1000	30	Electronics	Laptop
15-FEB-US	Sneakers	Nike	80	15	Fashion	Sneakers
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Coffee Maker
11-APR-US	smartphone	Samsung	900	25	Electronics	Smartphone
22-MAY-US	Backpack	North Face	70	20		Backpack
07-JUN-UK	Headphones	Sony		45	Electronics	Headphones
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	T-Shirt
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Blender
05-SEP-UK	Tablet	Apple	500	50	Electronics	Tablet
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Hiking Boots
17-JUN-IN	Laptop	HP	950	25	Electronics	Laptop
25-NOV-AU	Sneakers	Adidas	90	40		Sneakers
08-DEC-DE	Coffee Maker	Nespresso	120	35		Coffee Maker
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Smartwatch
16-APR-ES	headphones	Bose	250	20	Electronics	Headphones
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Laptop Bag
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Smartwatch
27-JAN-IT	laptop	Asus	980	10	Electronics	Laptop

- The Category column was cleaned using Find and Replace (CTRL + H) to maintain consistent naming across all entries.



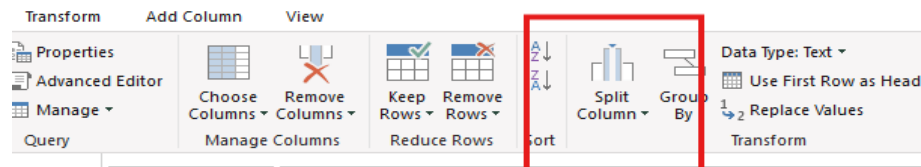
3. Splitting and Merging Data

3.1 Splitting Product Key

The **Product Key** column was split into:

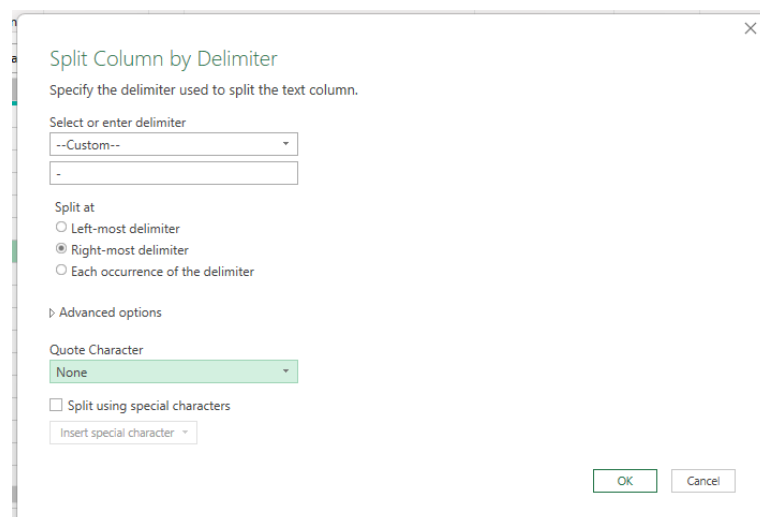
- Manufacturing Date
- Country Code

This was done using **Power Query** → **Split Column**.



The screenshot shows the Power Query ribbon with the 'Split Column' button highlighted. Below the ribbon, the formula bar displays the M code: `= Table.TransformColumnTypes(Source,{{"Product ID", type text}},`. The data table below has columns: Product ID, Product Name, Brand Name, and a blank column. The data rows are as follows:

	Product ID	Product Name	Brand Name	
1	28-JAN-US	laptop	Dell	
2	15-FEB-US	Sneakers	Nike	
3	03-MAR-US	Coffee Maker	Keurig	
4	11-APR-US	smartphone	Samsung	
5	22-MAY-US	Backpack	North Face	
6	07-JUN-UK	Headphones	Sony	
7	19-JUL-UK	T-shirt	Adidas	
8	23-AUG-UK	Blender	Ninja	
9	05-SEP-UK	Tablet	Apple	
10	14-OCT-UK	Hiking Boots	Timberland	
11	17-JUN-IN	Laptop	HP	
12	25-NOV-AU	Sneakers	Adidas	
13	08-DEC-DE	Coffee Maker	Nespresso	
14	18-FEB-CA	Smartwatch	Fitbit	
15	16-APR-ES	headphones	Bose	
16	21-AUG-CA	Laptop Bag	Samsonite	
17	20-AUG-CN	Smartwatch	Huawei	

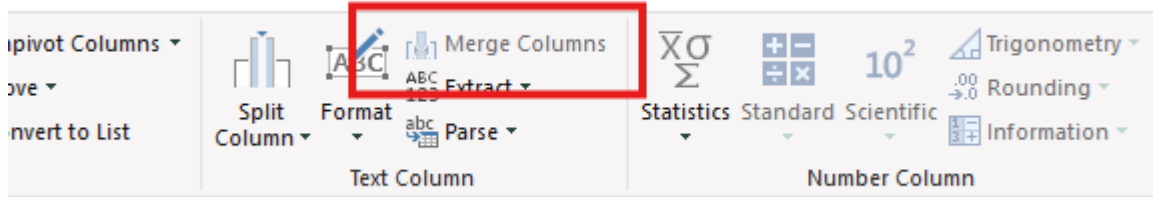


The 'Split Column by Delimiter' dialog box is shown. It has a title bar with a close button. The main text says 'Specify the delimiter used to split the text column.' Below this, there is a 'Select or enter delimiter' section with a dropdown menu set to '--Custom--' and a text input field containing '-'. The 'Split at' section has three radio buttons: 'Left-most delimiter', 'Right-most delimiter' (which is selected), and 'Each occurrence of the delimiter'. The 'Advanced options' section is expanded, showing a 'Quote Character' dropdown set to 'None', a checkbox for 'Split using special characters' which is unchecked, and an 'Insert special character' dropdown.

3.2 Merging Columns

The **Brand Name** and **Product Name** columns were merged to create a new field: **Product Brand**. A hyphen ("-") was used as the separator.

Product Name	Brand Name	
laptop	Dell	
Sneakers	Nike	
Coffee Maker	Keurig	
smartphone	Samsung	
Backpack	North Face	
Headphones	Sony	
T-shirt	Adidas	
Blender	Ninja	



Merge Columns

Choose how to merge the selected columns.

Separator

--Custom--
-

New column name (optional)

Merged

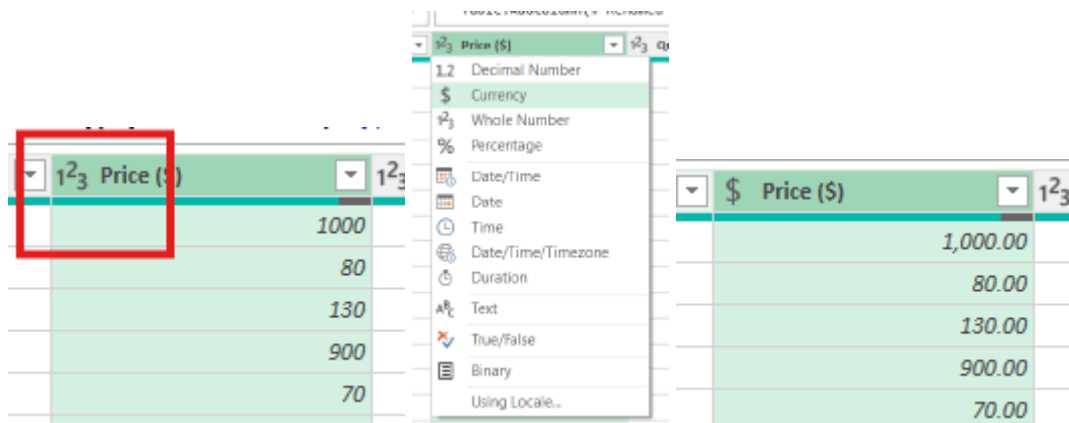
OK

Cancel

	H
	Product Category
s	laptop-Dell
	Sneakers-Nike
	Coffee Maker-Keurig
s	smartphone-Samsung
	Backpack-North Face
	Headphones-Sony
	T-shirt-Adidas
	Blender-Ninja
s	Tablet-Apple
	Hiking Boots-Timberland
s	Laptop-HP
	Sneakers-Adidas
	Coffee Maker-Nespresso

4. Number Formatting

- After splitting the Product Key in Power Query, the Manufacturing Date is now formatted as **DD-MM-YYYY**.
- The **Price** column was converted to **Currency format** using Power Query.



5. Handling Missing Values

Missing values were found in the **Price** and **Category** columns.

- Missing **Price** values were filled using the **average** of existing values.
- Missing **Category** values were replaced with “**Unknown.**”

```
=IF(ISBLANK(D2),AVERAGE(D2:D32),D2)
```

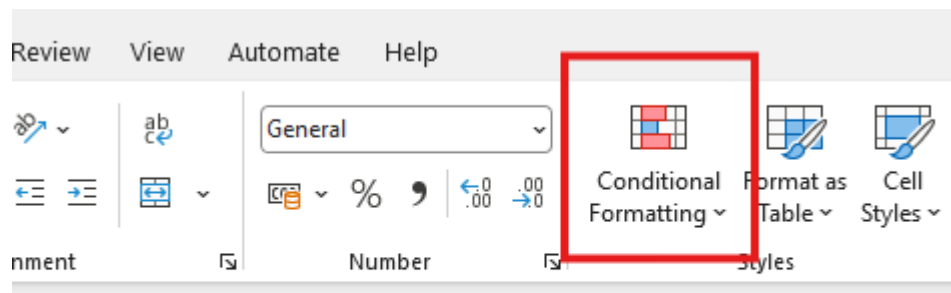
```
=IF(ISBLANK(F2),"Unknown",F2)
```

Price (\$)	Product Category
₹ 1,000.00	Electronics
₹ 80.00	Fashion
₹ 130.00	Kitchen
₹ 900.00	Electronics
₹ 70.00	Unknown
₹ 267.83	Electronics
₹ 30.00	Fashion
₹ 90.00	Kitchen
₹ 500.00	Electronics
₹ 130.00	Outdoor
₹ 950.00	Electronics
₹ 90.00	Unknown
₹ 120.00	Unknown
₹ 150.00	Electronics
₹ 250.00	Electronics
₹ 50.00	Accessories
₹ 160.00	Electronics

6. Conditional Formatting

To improve data visualization:

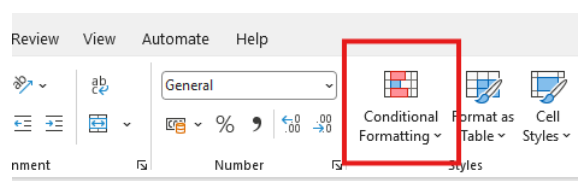
- **Data Bars** were applied to the **Price** column.
- A custom rule was created to highlight “**Electronics**” in the **Category** column using **Text That Contains** formatting.



	G	H	I	J	K	L
	Column1	Price (\$)	Product Category			

The screenshot shows the 'Conditional Formatting Rules Manager' dialog box. The 'Rules' list on the left contains a rule named 'Highlight Cells with Text' that uses the formula '=A1="Electronics"'. The 'Format' button is highlighted, and the 'Format Cells' dialog box is open, showing the 'Fill' tab with a light red fill color selected.

	G	H	I	J	K	L
	Column1	Price (\$)	Product Category			
		Price (\$)	Product Category			
		₹ 1,000.00	Electronics			
		₹ 80.00	Fashion			
		₹ 130.00	Kitchen			
		₹ 900.00	Electronics			
		₹ 70.00	Unknown			
		₹ 267.83	Electroni			
		₹ 30.00	Fashion			
		₹ 90.00	Kitchen			
		₹ 500.00	Electronics			
		₹ 130.00	Outdoor			
		₹ 950.00	Electronics			



	G	H	I	J	K	L
	Column1	Price (\$)	Product Category			

Text That Contains

Format cells that contain the text:

Electronics with Light Red Fill with Dark Red Text

OK Cancel

	I	J	K
	Price (\$)	Product Category	
	₹ 1,000.00	Electronics	
	₹ 80.00	Fashion	
	₹ 130.00	Kitchen	
	₹ 900.00	Electronics	
	₹ 70.00	Unknown	
	₹ 267.83	Electronics	
	₹ 30.00	Fashion	
	₹ 90.00	Kitchen	
	₹ 500.00	Electronics	
	₹ 130.00	Outdoor	
	₹ 950.00	Electronics	
	₹ 90.00	Unknown	